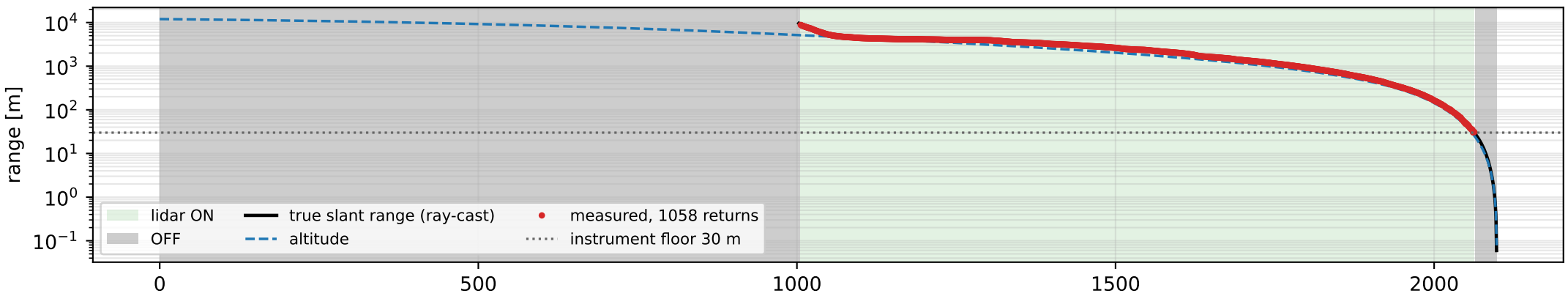
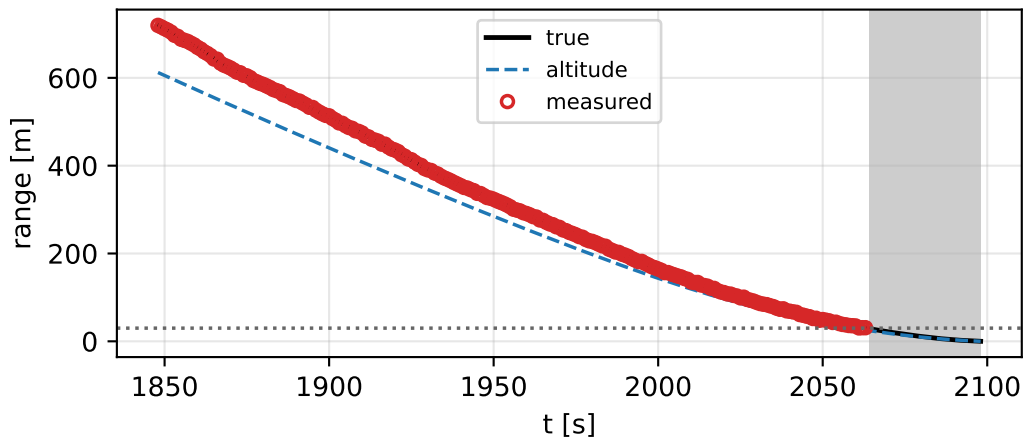
LIDAR block on the flown descent - closed_loop_imuplot.json
one beam along -thrust, Hayabusa2 LIDAR (Mizuno et al. 2017): 30 m - 25 km, 1-5.5 m, 0.5 m resolution, 1 Hz | incidence gate 70 deg, seed 1

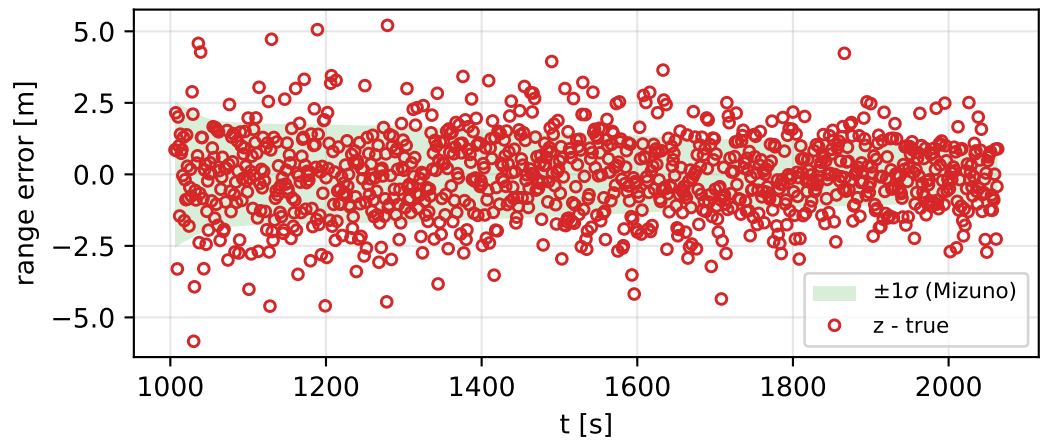
TRUE z (the slant range the beam actually travels) against the measurement, and against the altitude it is NOT



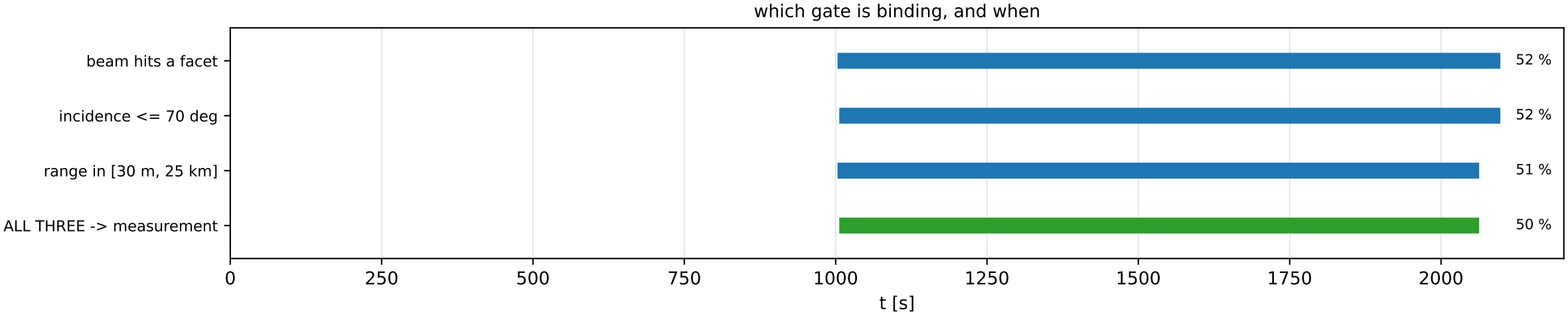
last 250 s: the beam goes dark below the 30 m floor



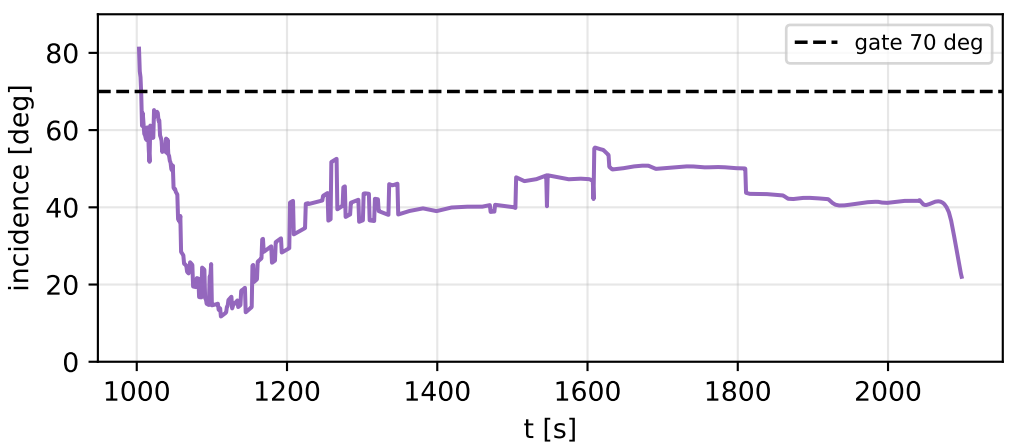
measurement error inside the range-dependent envelope



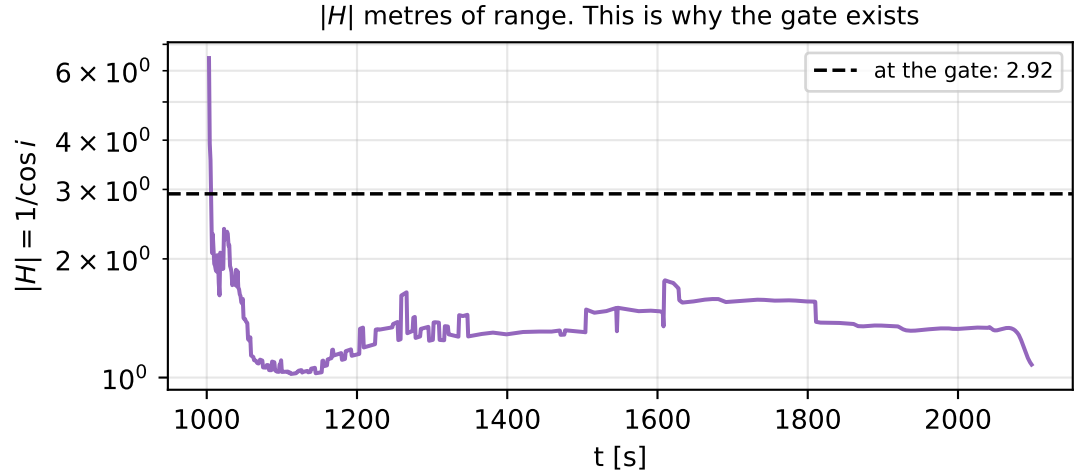
which gate is binding, and when



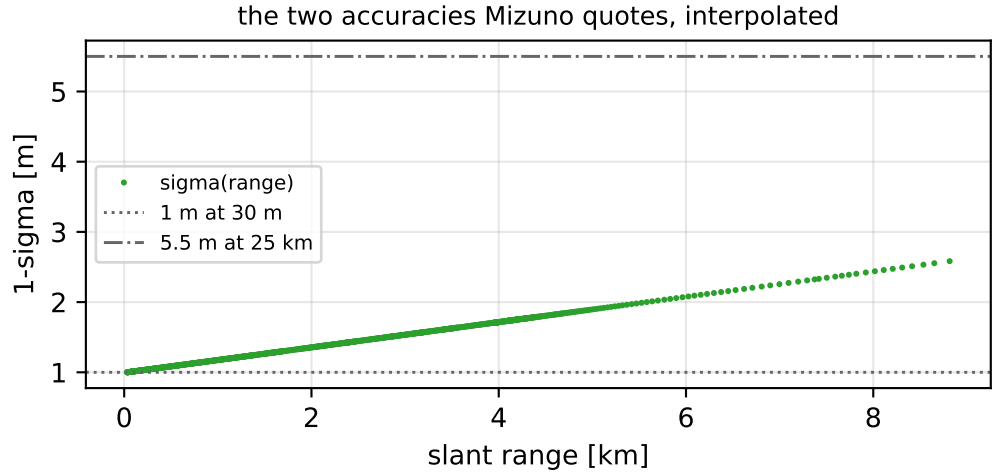
incidence of the beam on the facet it hits



range sensitivity to position: 1 m of position -> |H| metres of range. This is why the gate exists



the two accuracies Mizuno quotes, interpolated



TEST: 70 % within 1σ (expect 68 %), n = 1058
mean -0.060, std 0.994 (expect 0, 1)

